Assignment No.2

Aim

Develop a Python script to perform part-of-speech tagging on a short text dataset.

Objective

1. To understand the concept of Part-of-Speech (POS) tagging in Natural Language Processing.
2. To preprocess text data by cleaning, tokenizing.
3. To implement POS Tagging using Python libraries such as NLTK and SpaCy.
4. To evaluate tagging results on a sample dataset.

Expected Outcomes

1. Successful preprocessing of the input data includes tokenization and Normalization
2. Identification of Part-of-Speech and mapping of each word to its corresponding POS Tag (Noun , Verb , Adjective etc.).
3. Working Implementation of POS Tagging.
4. Better understanding of syntactic analysis in NLP

Theory

1.Introduction to Part-of-Speech Tagging

Part-of-Speech (POS) tagging is a fundamental task in Natural Language Processing (NLP) that involves assigning a grammatical category (tag) to each word in a sentence. These grammatical categories include noun, verb, adjective, adverb, pronoun, conjunction, etc.

Example : "The quick brown fox jumps over the lazy dog."

Tagged Output:

- The → Determiner (DT)
- quick → Adjective (JJ)
- brown → Adjective (JJ)
- fox → Noun (NN)
- jumps → Verb (VBZ)
- over → Preposition (IN)
- the → Determiner (DT)
- lazy → Adjective (JJ)
- dog → Noun (NN)

Thus, POS tagging helps in identifying the syntactic structure of a sentence

2. Parts of Speech and their Tags

Few common POS tags include:

Tag	Meaning
NN	**Noun**
VB	**Verb**
JJ	**Adjective**
RB	**Adverb**
PRP	**Pronoun**
DT	**Determiner**
IN	**Preposition**
CC	**Conjunction**

3. Why is POS Tagging Necessary?

English language has many homographs (i.e words with same spelling but different meaning) where the meaning of the word and grammatical role depends on the context.

Example: Word - "Book"

1) book (Noun) - I read a book.
2) book (Verb) - I will book a ticket.

This generates ambiguity in the understanding of language. This is known as Lexical Ambiguity. POS tagging helps to resolve lexical ambiguity using context.

4. Types of POS Tagging Approaches

1) **Rule-Based POS Tagging**
   Rule based tagging uses hand-written grammatical rules. It applies rules based on word suffixes or surrounding words.

   Example : If a word ends with "-ing", it is likely a verb (VBG).

   Advantages:
   - Easy to understand
   - No training data required

Disadvantages:

- Requires many complex rules
- Not scalable

2) Statistical (Probabilistic) POS Tagging

Uses probability models trained on annotated corpora.
Common models:

- Hidden Markov Model (HMM)
- Maximum Entropy Model

It calculates:

P(tag|previoustag)

Advantages:

- More accurate than rule-based
- Learns from real data

5. Pseudocode

```
START
INPUT: Text sentence or short text dataset

STEP 1: Import Required Libraries
      Import nltk
      Import word_tokenize from nltk.tokenize
      Import pos_tag from nltk

STEP 2: Download Required Resources (only once)
      Download 'punkt' tokenizer
      Download 'averaged_perceptron_tagger'

STEP 3: Read Input Text
      Store input sentence in a variable

STEP 4: Text Preprocessing
      1. Convert text to lowercase
      2. Remove unwanted characters
      3. Tokenize text using word_tokenize()
      4. Store tokens in a list

STEP 5: Apply POS Tagging
      Call pos_tag() function on token list
      Store output as (word, tag) pairs

STEP 6: Display Results
      For each (word, tag) pair:
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      Print word and its POS tag
OUTPUT:
     Word → POS Tag mapping
END
```

Results

```
POS Tagging Output:
Sentence 1 POS Tags: [('The', 'DET'), ('cat', 'NOUN'), ('is', 'AUX'), ('sleeping', 'VERB'), ('on', 'ADP'), ('the', 'DET'), ('mat', 'NOUN'), ('.', 'PUNCT')]
Sentence 2 POS Tags: [('Dogs', 'PROPN'), ('bark', 'NOUN'), ('loudly', 'ADV'), ('.', 'PUNCT')]
Sentence 3 POS Tags: [('He', 'PRON'), ('plays', 'VERB'), ('football', 'NOUN'), ('every', 'DET'), ('weekend', 'NOUN'), ('.', 'PUNCT')]
```

POS Tagging Output:

Sentence 1 POS Tags: [('The', 'DET'), ('cat', 'NOUN'), ('is', 'AUX'), ('sleeping', 'VERB'), ('on', 'ADP'), ('the', 'DET'), ('mat', 'NOUN'), ('.', 'PUNCT')]

Sentence 2 POS Tags: [('Dogs', 'PROPN'), ('bark', 'NOUN'), ('loudly', 'ADV'), ('.', 'PUNCT')]

Sentence 3 POS Tags: [('He', 'PRON'), ('plays', 'VERB'), ('football', 'NOUN'), ('every', 'DET'), ('weekend', 'NOUN'), ('.', 'PUNCT')]

Conclusion

Part-of-Speech tagging is a fundamental task in Natural Language Processing that identifies the grammatical role of words in a sentence. In this assignment, POS tagging was implemented using Python to analyze a short text dataset. The system successfully examined the sentence structure and assigned appropriate grammatical categories to each word. The implementation demonstrated the practical application of syntactic analysis and contributed to resolving lexical ambiguity.